UMANG ARORA

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MCA student who enjoys building reliable software applications through hands-on project work and real-world problem solving.

EDUCATION

Vellore Institute of Technology (VIT), Vellore *07/2025 – Present*
Master of Computer Applications (MCA) | CGPA (First Year): 8.48

Goswami Ganesh Dutta Sanatan Dharma College, Panjab University, Chandigarh *08/2022 – 07/2025*
Bachelor of Computer Applications (BCA) | Percentage: 73.87%

TECHNICAL SKILLS

Languages: Java, JavaScript, SQL
Frameworks & Libraries: Spring Boot, Node.js, React.js
Databases: SQL, MongoDB, Redis
Tools & Concepts: OOPs, Kafka, JWT, DSA, Collection Frameworks, REST APIs

PROJECTS

MINIT – Full-Stack Campus-Wide Commerce Platform *(Node.js, React, MongoDB, Kafka, Redis)* min.umangarora.in

→ Architected and developed a full-stack MERN marketplace enabling 50+ merchants and 5,000+ student users for campus-wide commerce with real-time order tracking and secure payments.
→ Integrated Apache Kafka for an event-driven architecture, processing 1,000+ concurrent events per second to manage real-time order updates and notifications.
→ Implemented Redis caching to optimize platform performance, reducing API latency for 30% of critical requests.

CodXp – Coding LMS Platform *(Spring Boot, MongoDB, JWT, Sandbox)*

→ Developed a role-based learning management system using Spring Boot and MongoDB, securing access with JWT authentication, impacting 5,000 daily active users.
→ Engineered an isolated execution sandbox using Docker to run untrusted code against automated test cases, preventing unauthorized system calls and resource exhaustion.
→ Designed an asynchronous submission grading queue that handles concurrent test case evaluations, achieving <200ms latency per sub-second test execution.

Differential Trust Engine – AI Brand Logo Trust Inference*(LightGBM, ResNet50, SHAP)* trustengine.umangarora.in

→ Built a context-adaptive XAI system converting brand logos into consumer trust scores via matched-pair differential analysis and a 745-palette human-curated emotion manifold; validated on 167,140 real-world logos ($R^2 = 0.73$).
→ Patented two core mechanisms — Achromatic Signal Gate and Dominant Signal Affirmation — enabling context aware SHAP calibration that replaces subjective design opinion with quantitative prescriptions.
→ Published as Indian Patent **IN202641096863 A1** through VIT's IPR & TT Cell.

LANGUAGES

→ **English** – Professional Proficiency
→ **Hindi** – Professional Proficiency

SOFT SKILLS

→ Team Coordination - Collaborated effectively with 5 team members to complete 2 projects successfully.
→ Solved 500+ quantitative aptitude problems, integrating data structures to strengthen analytical reasoning for complex software engineering challenges.